Questions-Gen — Three-Stage Training Architecture

Training corpus, SFT, GRPO, distillation, validation, and deployment interfaces

Dashed frame = architecture zone
Solid frame = subsystem
Box = module and responsibility

PROJECT OVERVIEW

Questions-Gen is a Qwen3-14B system for generating and evaluating mathematical competition problems.
The architecture contains the training corpus, SFT, GRPO, distillation, validation, and deployment interfaces.

TRAINING DATA · documented mixture

- **Historical** — 50% — QuestionDataPreparer.py
- **Conditional Variants** — 30% — data_preparer.py
- **Innovative Types** — 20% — real_math_data_processor.py

Stage 1 · SFT

- **Qwen3-14B** — base checkpoint
- **QuestionsGenTrainer.py** — src/ + questions_gen/core/trainer.py
- **stage1-fp-merged** — Hugging Face

Stage 2 · GRPO

RewardCalculator.py

- **Difficulty** — 0.4 difficulty
- **Novelty** — 0.3
- **Rigor** — 0.2
- **Diversity** — 0.1

Optimization

- **GRPO Groups** — 8 problems/group
- **NoveltyConstraint.py** — duplication guard
- **stage2-merged-16bit** — variation generation

Stage 3 · Distill

- **DeepSeekTeacher.py** — DeepSeek-R1
- **Teacher Signals** — difficulty · rigor
innovation · education
- **final-merged-16bit** — Hugging Face

VALIDATION · questions_gen/validation

- **model_validator.py** — single checkpoint
- **batch_validator.py** — algebra · calculus
geometry · number theory
- **quality_evaluator.py** — clarity · difficulty
originality

DEPLOYMENT · interfaces

- **main_cli.py** — validate · batch · quality
- **hf_utils.py** — three published stages
- **ollama_manager.py** — local inference
- **examples/** — quick_demo.py
demo_model_validation.py

SFT, GRPO, and distillation are three model stages. RewardCalculator and NoveltyConstraint belong to GRPO. Validation and deployment are separate interfaces.